

Unit 4

STATES OF MATTER

62 items · 211 marks

Syllabus sections in this unit

4.1 The gaseous state: ideal and real gases and $pV = nRT$

4.2 Bonding and structure

Unit 4: States of matter**1. M/J 26/P21/Q2/c(ii)**

2 The Contact process uses sulfur to produce sulfuric acid.

stage 1 Sulfur is burned in excess air.

stage 2 $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g}) \quad \Delta H = -196 \text{ kJ mol}^{-1}$

stage 3 SO_3 reacts to produce H_2SO_4 .

(c) CaCO_3 and O_2 are added to the waste gases from the Contact process before they escape. This prevents the release of SO_2 into the atmosphere.



(ii) Use the ideal gas equation to calculate the volume of $\text{O}_2(\text{g})$ that reacts with 0.500 mol of SO_2 at 400°C and 101 kPa in equation 1.

Include units in your answer.

volume of $\text{O}_2(\text{g}) =$

units

[4]

2. M/J 26/P23/Q2/b,c

2 Group 18 elements exist as colourless gases at room temperature and pressure.

Table 2.1

atom	relative volume of atom
neon	172
argon	287
krypton	355
xenon	446

(b) Identify **one** Group 18 element from Table 2.1 that behaves most like an ideal gas under standard conditions. Explain your answer.

[3]

(c) A sample of 1.775 g of gas **X** occupies 617.0 cm^3 at 300 K and 101 kPa.

Calculate the relative molecular mass, M_r , of **X** assuming that **X** behaves as an ideal gas.

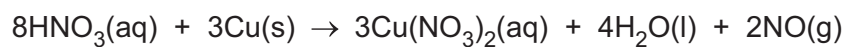
M_r of **X** =

[4]

3. MAR 26/P22/Q2/b

2 Nitrogen monoxide, NO, is an atmospheric pollutant.

- (b)** In the laboratory, NO is made in the reaction of copper metal with dilute nitric acid. The reaction is kept cold to prevent the oxidation of NO.



A student reacts 0.00393 mol Cu with excess dilute $\text{HNO}_3(\text{aq})$ at 5°C and 101 kPa.

Calculate the maximum volume, in cm^3 , of NO(g) that is produced under these conditions.

volume of NO(g) produced = cm^3 [4]

4. O/N 25/P24/Q2/a

2 Aluminium oxide, Al_2O_3 , and phosphorus(V) oxide, P_4O_{10} , are both used as reagents and catalysts.

- (a)** The melting point of Al_2O_3 is 2072°C . The melting point of P_4O_{10} is 340°C .

Explain the difference in the melting points of these two compounds.

[3]

5. M/J 25/P21/Q1/a(i,iii),d

- 1 (a)** Solid sodium conducts electricity. Sodium oxide conducts electricity when molten but not when solid.

(i) Name the type of bonding present in sodium and in sodium oxide.

bonding in sodium

bonding in sodium oxide

[1]

(iii) Explain why sodium oxide conducts electricity when molten.

[1]

(d) Table 1.2 shows melting points of some oxides.

Table 1.2

oxide	melting point/°C
SO ₂	−73
H ₂ O	0
SO ₃	17
SiO ₂	1610
MgO	2852
Al ₂ O ₃	2072

A student suggests the following hypothesis.

*The higher the oxidation number of the element combined with oxygen,
the higher the melting point of the oxide.*

Use Table 1.2 to deduce if this hypothesis is true or false or if there is **not** enough information to make a conclusion. Explain your answer.

[2]

6. M/J 25/P22/Q1/a,d

- 1 (a)** Diamond and graphite are both crystalline solids made from carbon atoms. Graphite conducts electricity. Diamond does **not** conduct electricity.

(i) Name the type of lattice structure present in the crystalline solids diamond and graphite.

[1]

(ii) Explain how graphite conducts electricity.

[1]

(iii) Explain why diamond does **not** conduct electricity.

[1]

(d) The melting points of different oxides are shown in Table 1.1.

Table 1.1

oxide	melting point/°C	force of attraction broken during melting
SO ₂	−73	
H ₂ O	0	
SiO ₂	1610	
Na ₂ O	1132	
MgO	2852	

- (i)** Complete Table 1.1 by identifying the strongest force of attraction in each oxide that is broken during melting. Use the abbreviations below.

i.d. = instantaneous dipole–induced dipole

p.d. = permanent dipole–permanent dipole

H = hydrogen bond

C = covalent bond

I = ionic bond

[3]

- (ii)** A student suggests the following hypothesis.

*The stronger the covalent bond between atoms in non-metal oxides,
the higher the melting point.*

Use Table 1.1 to deduce if this hypothesis is true or false or if there is **not** enough information to make a conclusion. Explain your answer.

[2]

7. M/J 25/P23/Q2/c(ii)

2 The chemical properties of oxides are related to the chemical bonding present in these compounds.

(c) Table 2.1 shows the melting points of some oxides.

Table 2.1

oxide	melting point/°C
SO ₂	−73
H ₂ O	0
SO ₃	17
SiO ₂	1610
Na ₂ O	1132
MgO	2852
Al ₂ O ₃	2072

(ii) A student suggests the following hypothesis.

The melting point of an ionically bonded oxide is only determined by the charge on the cation.

Use Table 2.1 to deduce if this hypothesis is true or false or if there is **not** enough information to make a conclusion. Explain your answer.

[2]

8. M/J 25/P24/Q1/a

1 Elements in Period 3 of the Periodic Table show trends in their properties.

(a) Complete Table 1.1 by identifying the lattice structures of the crystalline solids of Mg, Si and P.

Table 1.1

element	Mg	Si	P
lattice structure in crystalline solid			

[3]

9. M/J 25/P24/Q6/b

6 P, Q and R are three different hydrocarbon molecules.

(b) Q is a volatile hydrocarbon.

0.194 g of gaseous Q occupies a volume of 71.5 cm³ at 100 °C and 100 kPa.

Use the ideal gas equation to calculate the M_r of Q.

M_r of Q =

[2]

10. O/N 24/P21/Q3/a(i,ii),e(iii)

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

(a) Table 3.1 shows some properties of two Group 14 elements, C and Sn, in their standard states. The table is incomplete.

Table 3.1

	C (graphite)	Sn
state and appearance in standard state	grey shiny solid	silvery solid
electrical conductivity		good
type of bonding		metallic
type of structure	giant	

(i) Complete Table 3.1. [3]

(ii) Identify the lattice structure shown by graphite.

[1]

(e) Silicon readily reacts with elements of high electronegativity.

(iii) SiO_2 is a white solid that melts above 1700°C .

SiCl_4 is a colourless liquid at room temperature.

Explain the difference in the melting points of these two compounds with reference to their structure and bonding.

[2]

11. O/N 24/P22/Q3/d,e

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

(d) Complete Table 3.1 to show the properties of nitrogen and phosphorus in their standard states.

Table 3.1

	nitrogen	phosphorus
state and appearance of standard state	colourless gas	white solid
electrical conductivity		poor
type of bonding		
type of structure	simple	

[2]

(e) A form of solid nitrogen has a lattice structure similar to solid iodine.

Identify the type of lattice structure of solid nitrogen.

[1]

12. O/N 24/P23/Q3/a(i,ii),e(iii)

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

- (a)** Table 3.1 shows some properties of two Group 14 elements, C and Sn, in their standard states. The table is incomplete.

Table 3.1

	C (graphite)	Sn
state and appearance in standard state	grey shiny solid	silvery solid
electrical conductivity		good
type of bonding		metallic
type of structure	giant	

- (i)** Complete Table 3.1. [3]

- (ii)** Identify the lattice structure shown by graphite.

[1]

- (e)** Silicon readily reacts with elements of high electronegativity.

- (iii)** SiO_2 is a white solid that melts above 1700°C .

SiCl_4 is a colourless liquid at room temperature.

Explain the difference in the melting points of these two compounds with reference to their structure and bonding.

[2]

13. M/J 24/P21/Q2/d(ii)

- (d) Boron nitride is a white solid that melts above 2900 °C.

Fig. 2.2 shows part of the lattice structure of a crystal of boron nitride.

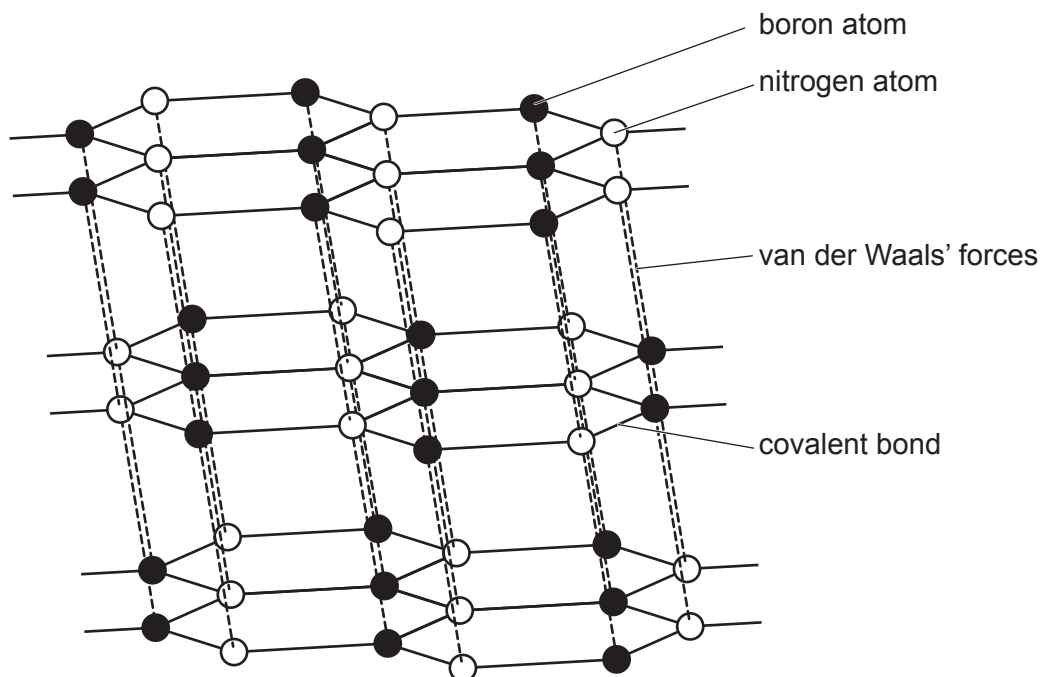


Fig. 2.2

- (ii) Suggest the identity of another crystalline solid that has atoms arranged in layers similar to that of solid boron nitride.

[1]

14. M/J 24/P23/Q3/c(i)

- (c) Boron carbide is a hard crystalline solid that has a melting point greater than 2000 °C.

- (i) Suggest the structure and bonding in boron carbide.

[1]

15. MAR 24/P22/Q1/b(ii)

- 1 Bismuth is an element in Group 15 of the Periodic Table.

- (b) Bismuth reduces water to form bismuth oxide, Bi_2O_3 . A colourless gas that ignites with a squeaky pop also forms.

- (ii) Bi_2O_3 is a yellow insoluble solid that melts at 1090 K. The molten compound conducts electricity.

Deduce the structure and bonding of Bi_2O_3 . Explain your answer.

[2]

16. O/N 23/P21/Q1/d(i)

- 1 The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

- (d) POC_l₃ shows similar chemical properties to PC_l₅.

POC_l₃ has a melting point of 1°C and a boiling point of 106°C.

POC_l₃ reacts vigorously with water, forming misty fumes and an acidic solution.

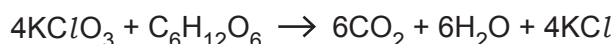
- (i) Explain how the information in (d) suggests the structure and bonding of POC_l₃ is simple covalent.

[2]

17. O/N 23/P21/Q3/d

- 3 Potassium chlorate, KClO₃, is widely used as an oxidising agent and to make O₂(g).

- (d) Molten KClO₃ reacts with glucose, C₆H₁₂O₆.



KClO₃ melts at 630 K. At this temperature, both CO₂ and H₂O are gases.

- (i) Use the ideal gas equation to calculate the volume, in m³, of one mole of gas at 630 K and 1.00 × 10⁵ Pa.

Show your working. Give your answer to 3 significant figures.

volume of 1 mol of gas = m³
[1]

- (ii) 5.00 g of C₆H₁₂O₆ reacts completely with molten KClO₃.

Use your answer to (d)(i) to calculate the total volume of gas released at 630 K and 1.00 × 10⁵ Pa in this reaction.

(If you were unable to answer (d)(i), use 0.0463 m³ in this question. This is **not** the correct answer to (d)(i).)

total volume of gas released = m³
[2]

18. O/N 23/P23/Q1/d(i)

- 1 The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

- (d) POC_l₃ shows similar chemical properties to PC_l₅.

POC_l₃ has a melting point of 1°C and a boiling point of 106°C.

POC_l₃ reacts vigorously with water, forming misty fumes and an acidic solution.

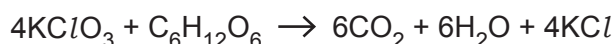
- (i) Explain how the information in (d) suggests the structure and bonding of POC_l₃ is simple covalent.

[2]

19. O/N 23/P23/Q3/d

- 3 Potassium chlorate, KClO₃, is widely used as an oxidising agent and to make O₂(g).

- (d) Molten KClO₃ reacts with glucose, C₆H₁₂O₆.



KClO₃ melts at 630 K. At this temperature, both CO₂ and H₂O are gases.

- (i) Use the ideal gas equation to calculate the volume, in m³, of one mole of gas at 630 K and 1.00 × 10⁵ Pa.

Show your working. Give your answer to 3 significant figures.

volume of 1 mol of gas = m³
[1]

- (ii) 5.00 g of C₆H₁₂O₆ reacts completely with molten KClO₃.

Use your answer to (d)(i) to calculate the total volume of gas released at 630 K and 1.00 × 10⁵ Pa in this reaction.

(If you were unable to answer (d)(i), use 0.0463 m³ in this question. This is **not** the correct answer to (d)(i).)

total volume of gas released = m³
[2]

20. M/J 23/P22/Q1/a(i,ii)

- 1 The melting points of some solids are shown in Table 1.1.

Table 1.1

solid	melting point/K
magnesium	923
phosphorus	317
sodium chloride	1074
sulfur	392

- (a) (i) State the type of bonding present in magnesium and in sodium chloride.

bonding in magnesium

bonding in sodium chloride

[1]

- (ii) Explain the difference in the melting points of magnesium and sodium chloride.

[1]

21. M/J 23/P23/Q1/a(i)

- 1 Copper is used in electrical equipment. It has a melting point of 1085 °C.

- (a) (i) Identify the lattice structure of copper.

[1]

22. MAR 23/P22/Q1/e(ii)

- 1 The Pauling electronegativity values of elements can be used to predict the chemical properties of compounds.

Use the information in Table 1.1 to answer the following questions.

Table 1.1

element	H	Li	C	O	S
Pauling electronegativity value	2.1	1.0	2.5	3.5	2.6
first ionisation energy/kJ mol ⁻¹	1310	519	1090	1310	1000
second ionisation energy/kJ mol ⁻¹	—	7300	2350	3390	2260

Table 1.2

compound	O=C=O	O=S=O	S=C=S	S=C=O
overall dipole moment				

- (ii) At 150 °C and 103 kPa, all of the compounds listed in Table 1.2 are gases.

Under these conditions, 0.284 g of one of the compounds occupies a volume of 127 cm³.

Use this information to calculate the M_r of the compound. Hence, identify the compound from those given in Table 1.2.

Show your working.

M_r =

identity of compound =

[3]

23. O/N 22/P22/Q2/d,f(ii)

- 2 The chlorides of some of the Period 3 elements are shown in Table 2.1.

Table 2.1

Period 3 chloride	NaCl	AlCl ₃	SiCl ₄	PCl ₅	PCl ₃	SCl ₂
bonding					C	C
structure					S	S
oxidation state of Period 3 element						

- (d) A 3.30 g sample of a Period 3 chloride is heated to 500 K in a sealed flask. At this temperature, the chloride is a gas of volume 250 cm³ and the pressure in the flask is 323 kPa.

Use the ideal gas equation $pV = nRT$ to calculate the M_r of the Period 3 chloride. Deduce its formula.

$$M_r =$$

formula of Period 3 chloride =

[3]

- (f) Bismuth is a dense metal in the same group as phosphorus.

- (ii) Bismuth reacts with chlorine to form BiCl₃.
BiCl₃ is a solid at room temperature. It melts when heated gently.
BiCl₃ reacts vigorously with water at room temperature to form an acidic solution.

Suggest the type of bonding and structure shown by BiCl₃. Explain your answer.

[2]

24. M/J 22/P22/Q1/a

- 1 (a) Magnesium has a melting point of 650 °C and high electrical conductivity.

Explain these properties of magnesium by referring to its structure and bonding.

[2]

25. M/J 22/P22/Q2/a

- 2 Radium, Ra, is an element found in Group 2 of the Periodic Table. It is a crystalline solid at room temperature and conducts electricity.

Radium chloride, RaCl₂, has a melting point of 900 °C and is soluble in water.

- (a) Predict the lattice structure of RaCl₂(s) based on the properties described.

[1]

26. M/J 22/P23/Q1/f(ii)

- (ii) Copper has a melting point of 1085 °C and a high electrical conductivity.

Explain these properties of copper by referring to its structure and bonding.

[2]

27. MAR 22/P22/Q2/c(i)

- 2 Some oxides of elements in Period 3 are shown.



- (c) P_4O_6 is a white solid that has a melting point of 24 °C. Solid P_4O_6 reacts with water to form H_3PO_3 .

- (i) Deduce the type of structure and bonding shown by P_4O_6 . Explain your answer.

[2]

28. O/N 21/P21/Q3/a(i)

- 3 Phosphorus is a reactive Period 3 element.

- (a) Phosphorus has several allotropes. Details of two allotropes are given.

allotrope of phosphorus	formula	melting point / °C
white	P_4	44
red	P	590

- (i) White phosphorus and red phosphorus both have covalent bonding.

Suggest the types of structure shown by white phosphorus (P_4) and red phosphorus (P).

Explain why red phosphorus (P) has a higher melting point than white phosphorus (P_4).

structure of P_4

structure of P

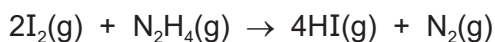
explanation

[3]

29. O/N 21/P22/Q1/e

- 1 Hydrogen iodide, HI, is a colourless gas at room temperature.

- (e) HI(g) can also be formed by the reaction of $\text{I}_2(\text{g})$ with hydrazine, $\text{N}_2\text{H}_4(\text{g})$.



State the change in pressure that would occur when 2 mol $\text{I}_2(\text{g})$ fully reacts with 1 mol $\text{N}_2\text{H}_4(\text{g})$ in a sealed container at constant temperature. Explain your answer.

[2]

30. O/N 21/P23/Q3/a(i)

3 Phosphorus is a reactive Period 3 element.

(a) Phosphorus has several allotropes. Details of two allotropes are given.

allotrope of phosphorus	formula	melting point/°C
white	P ₄	44
red	P	590

(i) White phosphorus and red phosphorus both have covalent bonding.

Suggest the types of structure shown by white phosphorus (P₄) and red phosphorus (P).

Explain why red phosphorus (P) has a higher melting point than white phosphorus (P₄).

structure of P₄

structure of P

explanation

[3]

31. M/J 21/P21/Q2/c,d

Data Booklet needed

2 Carbon monoxide gas, CO(g), and nitrogen gas, N₂(g), are both diatomic molecules.

(c) Both carbon monoxide and nitrogen are gases at room temperature and pressure.

They both behave like ideal gases under certain conditions.

(i) State the **two** conditions necessary for these two gases to approach ideal gas behaviour.

[1]

(ii) Explain why N₂(g) behaves more like an ideal gas than CO(g) does at 20.0 °C and 101 kPa.

[2]

(d) Calculate the amount, in mol, of pure nitrogen gas which occupies 100 cm³ at 101 kPa and 20.0 °C.

Use relevant information from the *Data Booklet*. Show your working.

Assume nitrogen behaves as an ideal gas.

mol
[3]

32. M/J 21/P22/Q2/a(ii),c(i,ii)

- 2** The strength of interaction between particles determines whether the substance is a solid, liquid or gas at room temperature.

(a) Lithium sulfide, Li_2S , is a crystalline solid with a melting point of 938°C . It conducts electricity when it is molten.

(ii) Explain, in terms of the structure of the crystalline solid, why lithium sulfide has a high melting point.

[2]

(c) Nitrogen, N_2 , is also a gas at room temperature and pressure. Neither CO nor N_2 is an ideal gas.

(i) State two assumptions that are made about the behaviour of particles in an ideal gas.

1

2

[2]

(ii) Explain why N_2 does not behave as an ideal gas at very high pressures.

[2]

33. MAR 21/P22/Q1/b

- 1** The rate of chemical reactions is affected by changes in temperature and pressure.

(b) The Boltzmann distribution assumes that the particles behave as an ideal gas.

(i) State **two** assumptions of the kinetic theory as applied to an ideal gas.

1

2

[2]

(ii) 2.00g of krypton gas, Kr(g) , is placed in a sealed 5.00 dm^3 container at 120°C .

Calculate the pressure, in Pa, of Kr(g) in the container.
Assume Kr(g) behaves as an ideal gas.

Show your working.

pressure = Pa [3]

(iii) State and explain the conditions at which krypton behaves most like an ideal gas.

[2]

34. MAR 21/P22/Q2/f(i)

2 Chlorine, Cl_2 , is a reactive yellow-green gas. It is a strong oxidising agent.

(f) 0.020 mol of element **Z** reacts with excess Cl_2 to form 0.020 mol of a liquid chloride.

The liquid chloride has formula ZCl_n , where n is an integer.

ZCl_n reacts vigorously with water at room temperature to give an acidic solution and a white solid.

When excess $\text{AgNO}_3(\text{aq})$ is added to the solution, 11.54 g of $\text{AgCl}(\text{s})$ forms.

(i) Suggest the type of bonding and structure shown by ZCl_n .

[1]

35. O/N 20/P21/Q2/a(iii)

2 Phosphorus, sulfur and chlorine can all react with oxygen to form oxides.

(a) Phosphorus reacts with an excess of oxygen to form phosphorus(V) oxide.

(iii) State the structure and bonding of solid phosphorus(V) oxide.

[1]

36. O/N 20/P22/Q2/d,f

2 The Period 3 elements, Na to S, all react with oxygen to form oxides.

(d) Explain why phosphorus(V) oxide has a low melting point of approximately 300°C but magnesium oxide has a high melting point of approximately 2850°C .

[3]

(f) Describe the lattice structure of silicon(IV) oxide.

Your answer should include reference to the arrangement of the silicon and oxygen atoms and the bonds between them.

[2]

37. O/N 20/P22/Q3/a(ii),d(iii)

3 PCl_5 , PCl_3 and NCl_3 are halides of Group 15 elements.

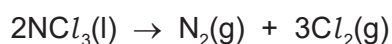
(a) PCl_5 can be formed from the reaction of phosphorus with chlorine. PCl_5 has a melting point of 161°C .

(ii) State the type of structure and bonding shown by liquid PCl_5 .

[1]

(d) NCl_3 is a yellow liquid that can be used to bleach flour.

(iii) $\text{NCl}_3(\text{l})$ decomposes according to the equation shown.



A sealed container of volume 250m^3 contains an unreactive gas at a pressure of $1.00 \times 10^5 \text{ Pa}$.

0.241g of $\text{NCl}_3(\text{l})$ was injected into the sealed container.

The sealed container was heated to make the $\text{NCl}_3(\text{l})$ decompose fully and then cooled to 20°C .

Calculate the final **total** pressure inside the sealed container at 20°C after the $\text{NCl}_3(\text{l})$ has fully decomposed.

final **total** pressure =

Pa

[4]

38. O/N 20/P23/Q2/a(iii)

2 Phosphorus, sulfur and chlorine can all react with oxygen to form oxides.

(a) Phosphorus reacts with an excess of oxygen to form phosphorus(V) oxide.

(iii) State the structure and bonding of solid phosphorus(V) oxide.

[1]

39. MAR 20/P22/Q1/b

1 Group 2 metals form alkaline solutions in water.

(b) The table shows the melting points of some Group 2 metal oxides.

compound	melting point/ $^\circ\text{C}$
MgO	2825
CaO	2613
SrO	2531
BaO	1923

Explain the trend in the melting points of the oxides down Group 2.

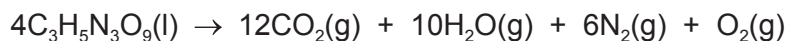
[2]

40. MAR 20/P22/Q3/b(ii)

3 Glycerol, $\text{CH}_2(\text{OH})\text{CH}(\text{OH})\text{CH}_2\text{OH}$, is widely used in the food industry and in pharmaceuticals.

(b) Glycerol can be used as a starting material in the manufacture of nitroglycerine, $\text{C}_3\text{H}_5\text{N}_3\text{O}_9$.

Nitroglycerine decomposes rapidly on heating to form a mixture of gases.



A sample of nitroglycerine decomposes, releasing 1.06 dm^3 of $\text{O}_2(\text{g})$ at 850 K and $1.00 \times 10^5 \text{ Pa}$.

(ii) Calculate the total volume of gas released by this decomposition at 850 K and $1.00 \times 10^5 \text{ Pa}$.

total volume of gas = dm^3 [1]

41. O/N 19/P21/Q2/b(i)

(b) (i) Explain why the boiling point of SiO_2 is much higher than the boiling point of SO_3 .

[3]

42. O/N 19/P22/Q1/b(ii)

1 In the Periodic Table, the p block contains elements whose outer electrons are found in the p subshell.

(b) Silicon is found in many compounds in the Earth's crust. Silicon has only three naturally occurring isotopes, ^{28}Si , ^{29}Si and ^{30}Si .

(ii) Silicon reacts with nitrogen gas to form Si_3N_4 .

Si_3N_4 is a solid with a melting point of 1900°C . It is insoluble in water and does not conduct electricity when molten.

Suggest the type of bonding in **and** structure of Si_3N_4 . Explain your answer.

[3]

43. O/N 19/P22/Q2/c

also Unit 3 (2 marks), Unit 2 (1 mark); this unit 2 marks

- 2** Oxygen is the most abundant element in the Earth's crust. It reacts with other elements to form stable compounds, ions and molecules.

(c) Sodium peroxide, Na_2O_2 , reacts with CO_2 .



The partial pressure of $\text{CO}_2(\text{g})$ in a 0.500 dm^3 sample of air is 5.37 kPa at 20°C .

- (i)** Calculate the amount, in moles, of $\text{CO}_2(\text{g})$ present in the sample of air at 20°C .

amount of $\text{CO}_2(\text{g}) =$ mol [2]

- (ii)** Calculate the mass of $\text{Na}_2\text{O}_2(\text{s})$ that would react fully with the amount of $\text{CO}_2(\text{g})$ calculated in **(i)**.

mass of $\text{Na}_2\text{O}_2(\text{s}) =$ g [1]

- (iii)** The peroxide ion, O_2^{2-} , has a single covalent bond between the two oxygen atoms. Each oxygen atom carries a negative charge.

Draw a 'dot-and-cross' diagram for the peroxide ion. Show outer electrons only.

[2]

44. O/N 19/P23/Q2/b(i)

- (b) (i)** Explain why the boiling point of SiO_2 is much higher than the boiling point of SO_3 .

[3]

45. M/J 19/P21/Q2/c

- 2** Magnesium silicide, Mg_2Si , is a compound made by heating magnesium with sand.

(c) Suggest, with reference to structure and bonding, why SiH_4 is a gas at room temperature.

[2]

46. M/J 19/P21/Q3/d

also Unit 2 (2 marks); this unit 5 marks

- (d) Chlorine exists as a diatomic gas, $\text{Cl}_2(\text{g})$. A sample of $\text{Cl}_2(\text{g})$ was made during a chemical reaction. When measured at 404 kPa and 25 °C the sample occupied a volume of 20.0 cm³.

- (i) Calculate the mass, in grams, of $\text{Cl}_2(\text{g})$ formed.

For this calculation, assume that chlorine behaves as an ideal gas under these conditions.

mass of $\text{Cl}_2(\text{g})$ = g [3]

- (ii) Calculate the number of chlorine atoms in this sample of $\text{Cl}_2(\text{g})$. You may find it helpful to use your answer to (d)(i).

If you are unable to calculate an answer to (d)(i), use 0.36g of Cl_2 . This is **not** the correct answer.

number of chlorine atoms = [2]

- (iii) $\text{Cl}_2(\text{g})$ does **not** behave as an ideal gas under these conditions.

Explain why $\text{Cl}_2(\text{g})$ behaves even **less** ideally at:

- very high pressures
- very low temperatures.

[2]

47. M/J 19/P23/Q2/a

- 2 Iodine is an element in Group 17 of the Periodic Table.

- (a) (i) At room temperature, iodine solid has a lattice structure.

Describe the arrangement of the iodine molecules within the solid.

[1]

- (ii) List **all** of the forces of attraction present in solid iodine and identify which of these are overcome when solid iodine is heated to produce iodine vapour.

force(s) of attraction present

force(s) of attraction overcome

[2]

Iodine reacts with aluminium to form a white solid, Al_2I_6 .

48. M/J 19/P23/Q3/c

- (c) Magnesium oxide is described as a ceramic material. It has a high melting point.

State and explain why ceramic materials such as magnesium oxide have high melting points.

[2]

49. MAR 19/P22/Q2/a(iii),d

also Unit 2 (1 mark); this unit 4 marks

- 2** The elements in Group 17 of the Periodic Table are called the halogens. They form stable compounds with both metals and non-metals.

The table gives some data about F_2 , HCl and CaF_2 .

	F_2	HCl	CaF_2
boiling point/K	85	188	2773
relative formula mass	38.0	36.5	78.1

- (iii)** Explain why CaF_2 has a very high boiling point.

[1]

- (d)** Interhalogen compounds, such as $BrCl$ or IF_5 , contain two or more different halogen atoms that are covalently bonded.

D is an interhalogen compound that contains only chlorine and fluorine.

At 0°C and $101\,325\text{ Pa}$, 1 dm^3 of **D** has a mass of 4.13 g .

- (i)** Use the general gas equation to calculate the relative molecular mass, M_r , of **D**.

$$M_r =$$

[3]

- (ii)** Use your answer to **(i)** to determine the molecular formula of **D**.

If you were unable to calculate the M_r in **(i)**, assume that the M_r is 130.5 . This is **not** the correct value.

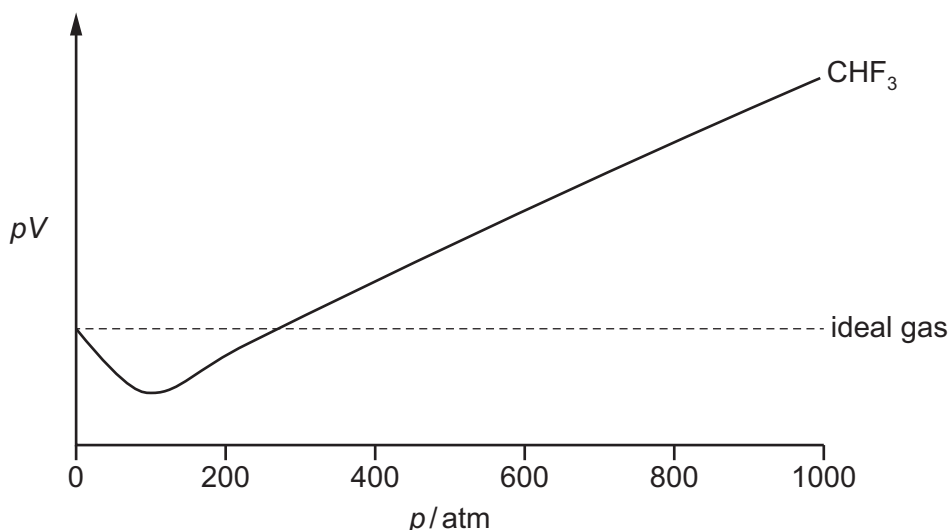
molecular formula of **D** =

[1]

50. O/N 18/P22/Q3/b

- 3** Trihalomethanes are organic molecules in which three of the hydrogen atoms of methane are replaced by halogen atoms, for example CHF_3 .

(b) The graph shows the relationship between pV and p at a given temperature for CHF_3 and an ideal gas.



- (i)** CHF_3 is not an ideal gas.

State **three** basic assumptions that scientists make about the properties of ideal gases.

- 1
- 2
- 3

[3]

- (ii)** Explain why CHF_3 deviates from the properties of an ideal gas at pressures greater than 300 atm.

[2]

51. M/J 18/P23/Q1/b

- 1** The elements sodium to chlorine, in the third period, all form oxides.

(b) (i) Explain why the melting point of MgO is higher than that of Na_2O .

[2]

(ii) Explain why the melting point of SiO_2 is much higher than that of SO_3 .

[2]

52. MAR 18/P22/Q2/a,d(i)

2 Carbon and silicon are elements in Group 14.

(a) C_{60} and diamond are allotropes of carbon.

(i) Describe the lattice structure of solid C_{60} .

[2]

(ii) C_{60} sublimes (turns directly from solid to gas) at about 800 K. Diamond also sublimes but only above 3800 K.

Explain why C_{60} and diamond sublime at such different temperatures.

[4]

(d) Silicon shows the same kind of bonding and structure as diamond.

(i) State the type of bonding and structure shown by silicon.

[2]

53. O/N 17/P22/Q1/b(i)

1 The elements sodium to sulfur react with chlorine. The melting points of some of the chlorides formed are shown.

chloride	$NaCl$	$MgCl_2$	$AlCl_3$	$SiCl_4$	PCl_3	SCl_2
melting point/K	1074	987	463	203	161	195

(b) (i) Explain, in terms of structure and bonding, why the melting point of $SiCl_4$ is much lower than that of $NaCl$.

[3]

54. M/J 17/P21/Q1/b(v)

- 1 Combustion data can be used to calculate the empirical formula, molecular formula and relative molecular mass of many organic compounds.

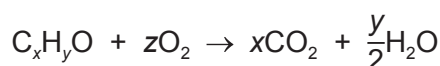
(b) **T** is an alcohol, C_xH_yO . A gaseous sample of **T** occupied a volume of 20 cm^3 at 120°C and 100 kPa .

The sample was completely burned in 200 cm^3 of oxygen (an excess). The final volume, measured under the same conditions as the gaseous sample, was 250 cm^3 .

Under these conditions, all water present is vaporised. Removal of the water vapour from the gaseous mixture decreased the volume to 170 cm^3 .

Treating the remaining gaseous mixture with concentrated alkali, to absorb carbon dioxide, decreased the volume to 110 cm^3 .

The equation for the complete combustion of **T** can be represented as shown.



- (v) Use the general gas equation to calculate the mass of **T** present in the original 20 cm^3 gaseous sample, which was measured at 120°C and 100 kPa .

Give your answer to **three** significant figures. Show your working.

mass = g [3]

55. M/J 17/P21/Q2/a

- 2 Structure and bonding can be used to explain many of the properties of substances.

(a) Copper, ice, silicon(IV) oxide, iodine and sodium chloride are all crystalline solids.

Complete the table with:

- the name of a type of bonding found in each crystalline solid,
- the type of lattice structure for each crystalline solid.

crystalline solid	type of bonding	type of lattice structure
copper		
ice		
silicon(IV) oxide		
iodine		
sodium chloride		

[5]

56. M/J 17/P23/Q1/b(v)

- 1 Combustion data can be used to calculate the empirical formula, molecular formula and relative molecular mass of many organic compounds. Combustion data cannot distinguish between different structural isomers.

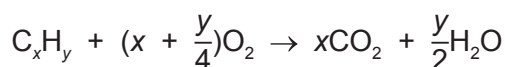
(b) **P** is a hydrocarbon, C_xH_y . A gaseous sample of **P** occupied a volume of 25 cm^3 at 37°C and 100 kPa .

The sample was completely burned in 200 cm^3 of oxygen (an excess).

The final volume, measured under the same conditions as the gaseous sample (so that the water produced is liquid and its volume can be ignored), was 150 cm^3 .

Treating the remaining gaseous mixture with concentrated alkali, to absorb carbon dioxide, decreased the volume to 50 cm^3 .

The equation for the complete combustion of **P** can be represented as shown.



- (v) Use the general gas equation to calculate the mass of **P** present in the original 25 cm^3 gaseous sample, which was measured at 37°C and 100 kPa .

Give your answer to **three** significant figures.

mass = _____ g [3]

57. MAR 17/P22/Q1/c(ii)

- (c) Cerium is a lanthanoid metal that shows similar chemical reactions to some elements in the third period. Most of cerium's compounds contain Ce^{3+} or Ce^{4+} ions.

(ii) Cerium(IV) oxide, CeO_2 , is a ceramic.

Suggest **two** physical properties of cerium(IV) oxide.

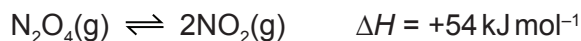
1.

2.

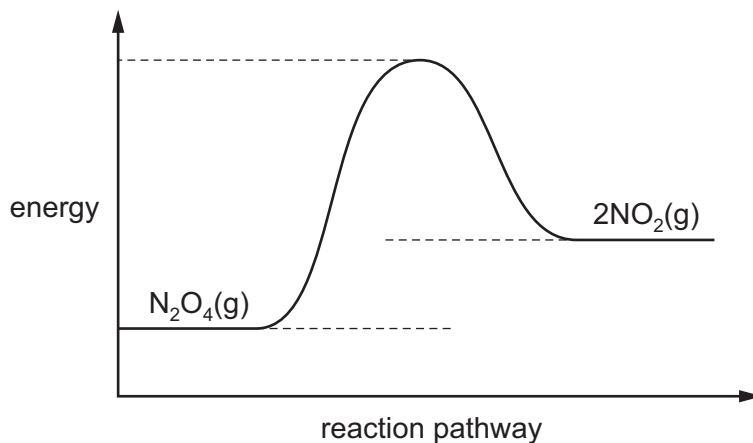
[2]

58. O/N 16/P22/Q2/b(i)

- 2 Dinitrogen tetraoxide, N_2O_4 , and nitrogen dioxide, NO_2 , exist in dynamic equilibrium with each other.



The energy profile for this reaction is shown.



- (b) 0.0500 mol of N_2O_4 was placed in a sealed vessel of volume 1.00 dm^3 , at a temperature of 50°C and a pressure of $1.68 \times 10^5 \text{ Pa}$. The mass of the resulting equilibrium mixture was 4.606 g.
- (i) Calculate the average molecular mass, M_r , of the resulting equilibrium mixture. Give your answer to **three** significant figures.

$M_r =$ [2]

59. M/J 16/P21/Q3/b(iii)

- 3 The elements in Group 2, and their compounds, show many similarities and trends in their properties.

(b) Magnesium oxide can be formed by the reaction of magnesium and oxygen in the air.

- (iii) Magnesium oxide is used to manufacture heat-resistant bricks for furnace linings in the steel-making industry.

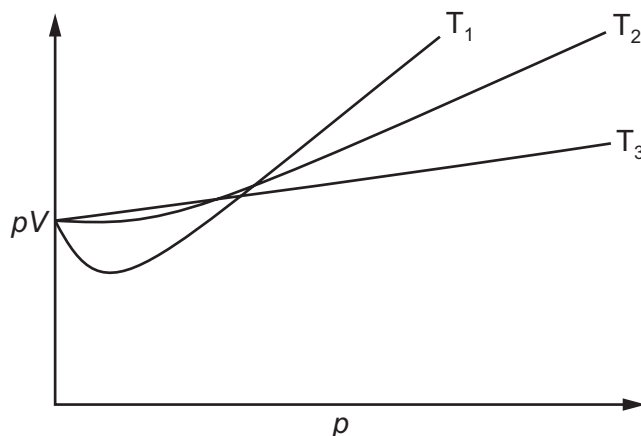
State and explain the property of magnesium oxide that makes it suitable for this use.

[2]

60. M/J 15/P21/Q2/a,b

- 2** The relationship $pV = nRT$ can be derived from the laws of mechanics by assuming ideal behaviour for gases.

- (a)** The graph represents the relationship between pV and p for a real gas at three different temperatures, T_1 , T_2 and T_3 .



- (i)** Draw **one** line on the graph to show what the relationship should be for the same amount of an **ideal** gas. [1]
- (ii)** State and explain, with reference to the graph, which of T_1 , T_2 or T_3 is the lowest temperature. [1]
- (iii)** Explain your answer to **(ii)** with reference to intermolecular forces. [1]
- (iv)** State and explain the effect of pressure on the extent to which a gas deviates from ideal behaviour. [2]
- (b)** A flask with a volume of 100 cm^3 was first weighed with air filling the flask, and then with another gas, **Y**, filling the flask. The results, measured at 26°C and $1.00 \times 10^5\text{ Pa}$, are shown.
- | | |
|-----------------------------------|------------------------------|
| Mass of flask containing air | = 47.930 g |
| Mass of flask containing Y | = 47.989 g |
| Density of air | = 0.00118 g cm^{-3} |
- Calculate the relative molecular mass, M_r , of **Y**.

M_r of **Y** = [4]

61. M/J 15/P23/Q1/d

1 Neon is a noble gas.

(d) A mixture of neon and argon has a mass of 0.275 g. The mixture was placed in a gas syringe at a temperature of 25 °C and a pressure of 100 kPa. Under these conditions the mixture was found to occupy a volume of 200 cm³.

(i) Calculate the average M_r of the mixture.

average M_r = [2]

(ii) Use your answer to **(i)** to calculate the percentage of neon in the mixture.
Give your answer to **three** significant figures.

percentage of neon = % [1]

62. M/J 15/P23/Q2/c

2 The elements in Group II, and their compounds, show a variety of trends in their properties.

(c) A refractory material is one that does not decompose or melt at very high temperatures. Over 50% of magnesium oxide production is for use as a refractory material.

Explain why magnesium oxide has a very high melting point.

[2]

Answers Section

1. M/J 26/P21/Q2/c(ii)

2(c)(ii)	M1 $PV=nRT$ M2 number of mol of O_2 required $n = 0.5 \div 2 = 0.25$ M3 temperature conversion to kelvin $= 400 + 273 = 673$ M4 $vol = (M2 \times 8.31 \times M3) \div 101\,000$ $= 0.0138431$ AND m^3	4 check the answer line first. M1 BOD upper / lower case discrepancies 0.0138(431) AND m^3 ✓✓✓✓ 13.8 and dm^3 OR 13 800 and cm^3 ✓✓✓✓ 0.014 and m^3 OR 14 and dm^3 or 14 000 and cm^3 ✓✓✓✓ 0.0276(86287) AND m^3 ✓✓✓ 0.00822(7722772) AND m^3 ✓✓✓✓ 0.0164(55445) AND m^3 ✓✓ A 2 sig figs up to calculator value final answers must be correctly rounded. M1–M3 may be seen in working if not described explicitly M1 $A V = nRT / P$ or A from working in M4 M3 A 273.15 OR 273.2 for 273 M3 A $400 + 273 / 673.15 / 673.2$ M4 volume calculated correctly using nRT / P AND correct / appropriate unit M4 A ecf from M2 and / or M3
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2. M/J 26/P23/Q2/b,c

2(b)	M1 neon / Ne (most like an ideal gas) M2 it has the smallest volume M3 weakest vdw / temporary dipole – induced dipole attractions / London (dispersion) forces / London forces	3 M2 and M3 comparative term required once only e.g. Ne has smaller volume and weak vdw M2 BOD 'it is the smallest (atom)' M2 I mass BOD molecules for atoms intermolecular forces / IMF's = vdw M3 BOD least for weakest M3 A 'forces of attraction between atoms' I smallest number of electrons
2(c)	M1 $PV=nRT$ M2 use of appropriate pressure value with volume e.g. convert volume to m^3 and kPa to Pa OR $101 \times 617.0 \times 10^{-3}$ etc. M3 calculate value for n $n = (101\,000 \times 617.0 \times 10^{-6}) \div 8.31 \times 300$ $= 0.024997 / 0.025$ M4 $(1.775 / M_r(X) = 0.02499 / 0.025)$ $M_r = 71(.0)$	4 Final answer 71(.0) ✓✓✓✓ M1 I subsequent incorrect rearrangement M2 A from calculations which produce consistent conversion for volume and pressure ie 101 and $617(.0) \times 10^{-3}$ (I ref to units for volume & pressure even if incorrect) M3 & M4 A ecf from incorrect conversion of volume and / or pressure units M3 A ecf from transcription errors / incorrect T using correct manipulation of $PV=nRT$ M4 A ecf from number of moles calculated M4 A ecf from M3 I units

3. MAR 26/P22/Q2/b

2(b)	M1 moles of NO = $0.00393 \times 2 / 3 = 0.00262$ M2 where $p = 101\,000$ (Pa) $R = 8.31$ $T = 278$ $n = 0.00262$ OR $p = 101$ (kPa) M3 use of $pV = nRT$ OR $V = nRT / p$ M4 volume = $5.99 \times 10^{-5} \times 10^6 = 59.9 \text{ cm}^3$ OR $5.99 \times 10^{-2} \times 10^3$	4
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4. O/N 25/P24/Q2/a

2(a)	M1 Al_2O_3 is ionic M2 P_4O_{10} is a (simple) molecule M3 ionic bonds are stronger than intermolecular forces	3
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5. M/J 25/P21/Q1/a(i,iii),d

1(a)(i)	sodium – metallic AND sodium oxide – ionic	1
1(a)(iii)	(sodium and oxide) ions move (through the liquid)	1
1(d)	<i>All three for two marks:</i> • false • comparison or statement of oxidation number • correct link to melting point from their comparison or statement	2

6. M/J 25/P22/Q1/a,d

1(a)(i)	(both) giant molecular	1															
1(a)(ii)	delocalised electrons move through the structure OR delocalised electrons move through / between the layers	1															
1(a)(iii)	no delocalised electrons (to carry charge through the structure) OR <u>all</u> (4) outer electrons are used in bonding	1															
1(d)(i)	<table border="1"> <tbody> <tr> <td>SO_2</td><td>–73</td><td>pd</td></tr> <tr> <td>H_2O</td><td>0</td><td>H</td></tr> <tr> <td>SiO_2</td><td>1600</td><td>C</td></tr> <tr> <td>Na_2O</td><td>1132</td><td>I</td></tr> <tr> <td>MgO</td><td>2852</td><td>I</td></tr> </tbody> </table>	SO_2	–73	pd	H_2O	0	H	SiO_2	1600	C	Na_2O	1132	I	MgO	2852	I	3
SO_2	–73	pd															
H_2O	0	H															
SiO_2	1600	C															
Na_2O	1132	I															
MgO	2852	I															
1(d)(ii)	<i>Evaluate hypotheses based on the information candidates have given in the table</i> M1 not enough information M2 SO_2 and / or H_2O break intermolecular forces OR SO_2 and / or H_2O does not break covalent bonds OR only SiO_2 breaks covalent bonds	2															

7. M/J 25/P23/Q2/c(ii)

2(c)(ii)	• false • Al_2O_3 melting point is lower than MgO • (but) charge on Al ion is greater than that on Mg Any 2 points = 1 mark , all 3 points = 2 marks	2
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8. M/J 25/P24/Q1/a

1(a)	<i>Mg</i>	<i>Si</i>	<i>P</i>	3
	M1 Giant metallic	M2 Giant molecular	M3 Simple molecular	

9. M/J 25/P24/Q6/b

6(b)	M1 convert temp to 373 K (and use) M2 correct rearrangement and use $7.15 = n \cdot 8.31(4) \times T$ to find n AND $M_r = 0.194 / n$ $M_r = 84.1$	2
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10. O/N 24/P21/Q3/a(i,ii),e(iii)

3(a)(i)	<table> <tr> <td>element</td><td>C (graphite)</td><td>Sn</td></tr> <tr> <td>state and appearance</td><td>grey shiny solid</td><td>silvery solid</td></tr> <tr> <td>electrical conductivity</td><td>• good/ conductor</td><td>good</td></tr> <tr> <td>type of bonding</td><td>• covalent</td><td>metallic</td></tr> <tr> <td>type of structure</td><td>giant</td><td>• giant</td></tr> </table>	element	C (graphite)	Sn	state and appearance	grey shiny solid	silvery solid	electrical conductivity	• good/ conductor	good	type of bonding	• covalent	metallic	type of structure	giant	• giant	3
element	C (graphite)	Sn															
state and appearance	grey shiny solid	silvery solid															
electrical conductivity	• good/ conductor	good															
type of bonding	• covalent	metallic															
type of structure	giant	• giant															
3(a)(ii)	giant molecular	1															
3(e)(iii)	M1 SiO ₂ is giant covalent AND SiCl ₄ is simple covalent / simple molecular / molecular M2 so need less energy to overcome IMF's in SiCl ₄ , compared to the bonds in SiO ₂ ora	2															

11. O/N 24/P22/Q3/d,e

3(d)	<table> <tr> <td>element</td><td>nitrogen</td><td>phosphorus</td></tr> <tr> <td>state and appearance</td><td>colourless gas</td><td>white solid</td></tr> <tr> <td>electrical conductivity</td><td>• poor</td><td>poor</td></tr> <tr> <td>type of bonding</td><td>• covalent</td><td>• covalent</td></tr> <tr> <td>type of structure</td><td>simple</td><td>• simple</td></tr> </table>	element	nitrogen	phosphorus	state and appearance	colourless gas	white solid	electrical conductivity	• poor	poor	type of bonding	• covalent	• covalent	type of structure	simple	• simple	2
element	nitrogen	phosphorus															
state and appearance	colourless gas	white solid															
electrical conductivity	• poor	poor															
type of bonding	• covalent	• covalent															
type of structure	simple	• simple															
3(e)	simple molecular (lattice structure)	1															

12. O/N 24/P23/Q3/a(i,ii),e(iii)

3(a)(i)	<table> <tr> <td>element</td><td>C (graphite)</td><td>Sn</td></tr> <tr> <td>state and appearance</td><td>grey shiny solid</td><td>silvery solid</td></tr> <tr> <td>electrical conductivity</td><td>• good/ conductor</td><td>good</td></tr> <tr> <td>type of bonding</td><td>• covalent</td><td>metallic</td></tr> <tr> <td>type of structure</td><td>giant</td><td>• giant</td></tr> </table>	element	C (graphite)	Sn	state and appearance	grey shiny solid	silvery solid	electrical conductivity	• good/ conductor	good	type of bonding	• covalent	metallic	type of structure	giant	• giant	3
element	C (graphite)	Sn															
state and appearance	grey shiny solid	silvery solid															
electrical conductivity	• good/ conductor	good															
type of bonding	• covalent	metallic															
type of structure	giant	• giant															
3(a)(ii)	giant molecular	1															

3(e)(iii)	M1 SiO ₂ is giant covalent AND SiCl ₄ is simple covalent / simple molecular / molecular M2 so need less energy to overcome IMF's in SiCl ₄ , compared to the bonds in SiO ₂ ora	2
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13. M/J 24/P21/Q2/d(ii)

2(d)(ii)	graphite	1
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14. M/J 24/P23/Q3/c(i)

3(c)(i)	giant (molecular and) covalent	1
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15. MAR 24/P22/Q1/b(ii)

1(b)(ii)	giant AND has a high melting point	1
	ionic AND conducts electricity (only) when molten	1

16. O/N 23/P21/Q1/d(i)

1(d)(i)	(relatively low melting and boiling points suggests) weak intermolecular / VdW forces	1
	(vigorous reaction with water suggests) hydrolysis	1

17. O/N 23/P21/Q3/d

3(d)(i)	$pV = nRT \therefore V = 8.31 \times 630 \div 100\,000 = \underline{\underline{0.0524}} \text{ (m}^3\text{)}$	1
3(d)(ii)	moles of gas = $12 \times 5.00 / 180$	1
	volume of gas = $M1 \times \text{(d)(i)} = \underline{\underline{0.0175}} \text{ (m}^3\text{)}$	1

18. O/N 23/P23/Q1/d(i)

1(d)(i)	(relatively low melting and boiling points suggests) weak intermolecular / VdW forces	1
	(vigorous reaction with water suggests) hydrolysis	1

19. O/N 23/P23/Q3/d

3(d)(i)	$pV = nRT \therefore V = 8.31 \times 630 \div 100\,000 = \underline{\underline{0.0524}} \text{ (m}^3\text{)}$	1
3(d)(ii)	moles of gas = $12 \times 5.00 / 180$	1
	volume of gas = $M1 \times \text{(d)(i)} = \underline{\underline{0.0175}} \text{ (m}^3\text{)}$	1

20. M/J 23/P22/Q1/a(i,ii)

1(a)(i)	bonding in magnesium – metallic AND bonding in sodium chloride – ionic	1
1(a)(ii)	bonds in NaCl are stronger than bonds in Mg	1

21. M/J 23/P23/Q1/a(i)

1(a)(i)	giant metallic	1
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22. MAR 23/P22/Q1/e(ii)

1(e)(ii)	conversion of units	103000 Pa	$127 \times 10^{-6} \text{ m}^3$	423 K	1
	Use of $pV = (m / M_r)RT$	$M_r = \frac{0.284 \times 8.31 \times 423}{103000 \times 127 \times 10^{-6}}$			1
	$M_r = 76.3$ AND compound = CS ₂				1

23. O/N 22/P22/Q2/d,f(ii)

2(d)	M1 value of n based on re-arranged equation for 'n' and substitution of correct values $n = \frac{3.23 \times 10^5 \cdot 250 \times 10^{-6}}{8.31 \cdot 500} = 0.0194(34416)$	1
	M2 calculated value of M_r based on M1 $M_r = 3.30 \div n = 170$	1
	M3 period 3 chloride which matches M_r in M2 (formula) SiCl ₄	1
2(f)(ii)	M1 (structure =) simple/molecular, because it has a low melting/boiling point owtte	1
	M2 (bonding =) covalent, because it is hydrolysed	1

24. M/J 22/P22/Q1/a

1(a)	M1 comment explaining high melting point of Mg in terms of many strong metallic bonds OR many strong (electrostatic) attractions between cations and delocalised electrons OR strong bonds in giant metallic structure. M2 comment explaining electrical conductivity of Mg in terms of movement of delocalised electrons delocalised electrons can move through the structure	2
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25. M/J 22/P22/Q2/a

2(a)	giant ionic	1
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26. M/J 22/P23/Q1/f(ii)

1(f)(ii)	M1 comment explaining high melting point of Cu many strong metallic bonds OR many strong (electrostatic) attractions between cations and delocalised electrons OR strong bonds in giant metallic structure. M2 comment explaining electrical conductivity of Cu delocalised electrons are free are to move through the structure (owtte)	2
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27. MAR 22/P22/Q2/c(i)

2(c)(i)	(structure =) simple/molecular, because it has a low melting/boiling point [1] (bonding =) covalent, because it is hydrolysed [1]	2
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28. O/N 21/P21/Q3/a(i)

3(a)(i)	M1 simple molecular M2 giant molecular M3 weak IMFs (overcome) in P ₄ AND strong (covalent) bonds (broken) in P	3
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29. O/N 21/P22/Q1/e

1(e)	M1 pressure increases M2 (pressure goes up as) number of moles/molecules increases in ratio 3 (gas) reactants to 5 (gas) products OR pressure is (directly) proportional to number of moles/molecules	2
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30. O/N 21/P23/Q3/a(i)

3(a)(i)	M1 simple molecular M2 giant molecular M3 weak IMFs (overcome) in P ₄ AND strong (covalent) bonds (broken) in P	3
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31. M/J 21/P21/Q2/c,d

2(c)(i)	high temperature AND low pressure	1
2(c)(ii)	M1 CO is polar / has a permanent dipole OR N ₂ is non-polar	1
	M2 IMF in CO are (more) significant / larger OR IMF in N ₂ are smaller / less significant	1
	<i>Alternative answer</i> M1 (Size of) N ₂ smaller than CO OR volume of N ₂ molecules / particles smaller	
	<i>Alternative answer</i> M2 volume of N ₂ molecules / particles is more negligible OR A	
2(d)	M1 correct conversion to consistent units P = 101 000 V = 100 / 1 000 000 = (1 × 10 ⁻⁴) T = 293	1
	M2 use of all values from M1 in correct relationship, n = PV / RT	1
	M3 calculation = 4.15 × 10 ⁻³ mol	1

32. M/J 21/P22/Q2/a(ii),c(i,ii)

2(a)(ii)	M1 giant	1
	M2 (many) strong force(s) of attraction between oppositely charged ions OR (many) strong ionic bond(s)	1
2(c)(i)	<i>Any two assumptions about the behaviour of particles in an ideal gas from</i> (particles / molecules have mass but) negligible size / volume (compared to total volume of gas / container) - no / negligible forces / interactions (between particles / molecules) - collisions are elastic	2
2(c)(ii)	M1 IMF become larger / more significant	1
	M2 volume of <u>molecules</u> / <u>particles</u> becomes significant / no longer negligible	1

33. MAR 21/P22/Q1/b

1(b)(i)	Any two from: - no VdW forces present / no forces of attraction between particles (ideal gas) particles have no / negligible volume (compared to container) - collisions between (ideal gas) particles / walls of container are perfectly elastic (ideal gas) particles behave as rigid spheres	2
1(b)(ii)	M1: moles of krypton = 2.00 ÷ 83.8 (= 0.0239 mol) M2: conversion of value into consistent units for $pV = nRT$ M3: $p = \frac{M1 \times 8.31 \times 393}{5.00 \times 10^{-3}} = 15600 \text{ Pa}$	3

1(b)(iii)	M1: low pressure AND high temperature M2: Either of: • volume of particles is negligible (compared to volume of container) • VdW forces are insignificant (owing to high kinetic energy of particles)	2
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34. MAR 21/P22/Q2/f(i)

2(f)(i)	simple / molecular AND covalent	1
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35. O/N 20/P21/Q2/a(iii)

2(a)(iii)	Simple and covalent OR molecular and covalent	1
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36. O/N 20/P22/Q2/d,f

2(d)	M1: <i>identification of forces broken during melting of phosphorus(V) oxide</i> intermolecular forces in phosphorus(V) oxide (are broken) M2: <i>identification of force broken during melting of magnesium oxide</i> electrostatic forces of attraction between (many oppositely charged) ions in magnesium oxide M3: <i>statement linking difference in strength of appropriate forces described in M1 & M2 to explain difference in melting point</i> (only) intermolecular forces weaker than forces (of attraction) between ions / ionic bonds	3
2(f)	M1: giant M2: covalent AND tetrahedral / four Si—O bonds	2

37. O/N 20/P22/Q3/a(ii),d(iii)

3(a)(ii)	simple / molecular AND covalent	1
3(d)(iii)	method 1 M1: <i>show the number of mol gas produced from 0.241 g NCl₃</i> <i>[M_r(NCl₃) = 120.5 1 mol NCl₃ produces 2 mol gas]</i> $n = 2 \times 0.241 / 120.5$ OR $n = 0.0040$ (mol gas produced.) M2: <i>correct conversion of T to Kelvin, V to m³ and correct value of R</i> M3: <i>use of equation $P = nRT / V$ with M2 values for n, R, T and V to find pressure of mol gas produced</i> increase in $p = nRT / V = \frac{0.0040 \times 8.31 \times 293}{250 \times 10^{-6}}$ $= 3.90 \times 10^4$ M4: $= 1.00 \times 10^5 + M3$ (Pa) total pressure $= 1.00 \times 10^5 + 3.90 \times 10^4$ $= 1.39 \times 10^5$ (Pa) method 2 M1: <i>calculate the produced from 0.241 g NCl₃</i> 0.003 (mol) Cl ₂ AND 0.001 (mol) N ₂ M2: <i>conversion of T to Kelvin, V to m³ and correct value of R in all PV/RT equations used</i> M3: <i>use of equation $PV / RT = n$ for both calculations or a combined equation with M2 values for R, T and V to find partial pressure for each of the gases</i> $pp_{Cl_2} = 29217.96$ AND $pp_{N_2} = 9739.32$ M4: <i>Use $P_{total} = pp_{unreacted\ gas} + pp_{Cl_2} + pp_{N_2}$</i> $1 \times 10^5 + 2.92 \times 10^4 + 9.74 \times 10^3 = 138\,940$	4

38. O/N 20/P23/Q2/a(iii)

2(a)(iii)	Simple and covalent OR molecular and covalent	1
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39. MAR 20/P22/Q1/b

1(b)	M1 (decreasing melting point down the group because) lower forces of attraction / weaker bonds (between cations and anions / oxide / O^{2-}) M2 larger cations and constant charge OR decreasing charge density of cation (down group)	2
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40. MAR 20/P22/Q3/b(ii)

3(b)(ii)	$1.06 \times 29 = 30.7(4) \text{ dm}^3$	1
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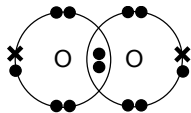
41. O/N 19/P21/Q2/b(i)

2(b)(i)	M1 SiO_2 has a network of strong bonds / SiO_2 has many strong bonds M2 SO_3 has weak intermolecular forces OR weak VdW forces (between molecules) M3 high(er) / more energy required to break bonds than overcome forces (between molecules)	3
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42. O/N 19/P22/Q1/b(ii)

1(b)(ii)	M1 giant (molecule) M2 strong covalent bonds (between atoms / particles) M3 no mobile charged particles / carriers	3
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43. O/N 19/P22/Q2/c

2(c)(i)	M1: correct conversion of quantities $V = 5(.00) \times 10^{-4} \text{ (m}^3\text{)}$ $T = 293 \text{ (K)}$ $p = 5.37(0) \times 10^3 \text{ (Pa)}$ M2: calculation to find n using $n = PV / RT$ $(n) = 1.1 \times 10^{-3} \text{ (mol)}$	2
2(c)(ii)	$(i) \times 78 = 0.0860 \text{ g}$	1
2(c)(iii)	 M1: bonding pair between the two O M2: total of 14 electrons distributed equally between the two O	2

44. O/N 19/P23/Q2/b(i)

2(b)(i)	M1 SiO_2 has a network of strong bonds / SiO_2 has many strong bonds M2 SO_3 has weak intermolecular forces OR weak VdW forces (between molecules) M3 high(er) / more energy required to break bonds than overcome forces (between molecules)	3
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45. M/J 19/P21/Q2/c

2(c)	M1 simple (covalent) / molecular / molecules	1
	M2 weak IMF / (temporary) induced dipole (forces)	1

46. M/J 19/P21/Q3/d

3(d)(i)	M1 correct conversions of data to SI/consistent units $p = 404\,000$; $V = 20 \times 10^{-6}$; $T = 298$		1
	M2 calculation of n ($= pV/RT$) from M1 values $n = \frac{404000 \times 20 \times 10^{-6}}{8.31 \times 298} = 3.263 \times 10^{-3} \text{ mol of Cl}_2$		1
	M3 finding the mass of Cl_2 $= 3.263 \times 10^{-3} \times 71.0 = 0.23 \text{ (g)}$		1
3(d)(ii)	Method 1 M1 $= 3.263 \times 10^{-3} \times 2$	Method 2 M1 $= \frac{0.23}{71.0} \times 2$ OR 6.53×10^{-3}	1
	M2 $= 6.02 \times 10^{23} \times \text{M1}$ $= 3.93 \times 10^{21}$ atoms of Cl	M2 $= 6.02 \times 10^{23} \times \text{M1}$ $= 3.90 \times 10^{21}$ atoms of Cl	1
3(d)(iii)	M1 size / volume of molecule / particle becomes significant / non-negligible OR IMFs become significant / non-negligible		1
	M2 IMFs becomes significant / non-negligible / collisions are not elastic		1

47. M/J 19/P23/Q2/a

2(a)(i)	held in regular / uniform arrangement	1
2(a)(ii)	M1 covalent (bonds) AND (temporary) induced dipoles M2 (temporary) induced dipoles	2

48. M/J 19/P23/Q3/c

3(c)	M1 giant (structure / lattice) M2 (so) lots of energy needed to break the bonds OR strong bonds	2
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49. MAR 19/P22/Q2/a(iii),d

2(a)(iii)	strong (electrostatic) forces of attraction between (oppositely charged) ions	1
2(d)(i)	$pV = nRT \therefore n = \frac{pV}{RT} = \frac{101\,325 \times 0.001}{8.31 \times 273} = 0.0447 \text{ mol}$ $\therefore M_r = \frac{m}{n} = \frac{4.13}{0.0447} = 92.4 \text{ or } 92.5$ M1 Use of $T = 273 \text{ K}$, $V = 0.001 \text{ m}^3$ and $p = 101325 \text{ Pa}$ M2 correct use of $pV = nRT$ using values from M1 M3 correct calculation of M_r using $4.13 \div \text{moles from M2}$	3
2(d)(ii)	ClF_3	1

50. O/N 18/P22/Q3/b

3(b)(i)	1 mark for each bullet, max 3 - particles / molecules have (mass but) negligible size / volume (compared to total volume of gas / container) - no / negligible forces / interactions between particles / molecules - collision between particles / molecules are elastic - gas obeys (all) basic gas laws	3
3(b)(ii)	M1 particles / molecules are (so) close [1] M2 particle / molecule size becomes significant [1] OR <u>repulsive</u> forces between particle / molecules become significant	2

51. M/J 18/P23/Q1/b

1(b)(i)	stronger attraction for O^{2-} / stronger ionic bonding / more energy needed to overcome ionic bonding / separate ions	1
	charge density of magnesium (ion) is greater (than sodium ion) ora	1
1(b)(ii)	Either: M1 SiO_2 has a giant (covalent) structure / giant molecular M2 Covalent bonds (much) stronger than VdW / id-id / IMFs in SO_3 OR M1 SO_3 has a (simple) molecular structure / (simple) molecule. M2 VdW / id-id / IMFs M2 IMF's are (much) weaker than covalent bonds (broken in SiO_2) OR M1 Covalent bonds are broken in SiO_2 AND VdW / id-id / IMFs in SO_3 M2 Covalent bonds are stronger (than VdW / id-id / IMFs)	2

52. MAR 18/P22/Q2/a,d(i)

2(a)(i)	simple molecular regular arrangement (of C_{60} molecules)	2
2(a)(ii)	C_{60} has (weak) intermolecular / VdW / London / dispersion / id-id forces (and covalent bonds) diamond has covalent bonds (diamond's) bonds are stronger more energy required / lots of energy to break (covalent bonds in diamond)	4
2(d)(i)	giant (molecular) (each Si has four) covalent (bonds)	2

53. O/N 17/P22/Q1/b(i)

1(b)(i)	$SiCl_4$ simple / molecular AND Van der Waals' / id-id forces / London / dispersion forces / IMFs	1
	$NaCl$ ionic OR giant	1
	bonding (in $NaCl$) stronger (than forces in $SiCl_4$) owtte	1

54. M/J 17/P21/Q1/b(v)

1(b)(v)	correct conversions of data to SI/consistent units $p = 100\,000$; $V = 20 \times 10^{-6}$; $T = 393$	1
	calculation of n ($= pV/RT$) from M1 values $n = \frac{100 \times 10^3 \times 20 \times 10^{-6}}{8.31 \times 393}$	1
	calculation of mass m ($= n \times Mr$) AND answer correct to 3sf $m = 6.12 \times 10^{-4} \times 60 = 0.0367$ (g)	1
	Alternative answer for using $C_4H_{10}O$: $m = 6.12 \times 10^{-4} \times 74 = 0.0453$ (g)	

55. M/J 17/P21/Q2/a

2(a)	substance	type of bonding	type of lattice structure	
	copper	metallic	giant/metallic	1
	ice	covalent OR hydrogen(-bonding) / H(-bonding)	hydrogen-bonded / simple / molecular	1
	silicon(IV) oxide	covalent	giant (molecular) / macromolecular	1
	iodine	covalent	simple / molecular	1
	sodium chloride	ionic	giant / ionic	1

56. M/J 17/P23/Q1/b(v)

1(b)(v)	correct conversions of data to SI / consistent units $P = 100\,000$; $V = 25 \times 10^{-6}$; $T = 310$	1
	calculation of n ($= pV / RT$) $n = \frac{100 \times 10^3 \times 25 \times 10^{-6}}{8.31 \times 310}$	1
	calculation of mass m ($= n \times M_r$) AND answer correct to 3sf $m = 9.705 \times 10^{-4} \times 56 = 0.0543$ (g)	1
	Alternative answer for using C_6H_{12} : $m = 9.705 \times 10^{-4} \times 84 = 0.0815$ (g)	

57. MAR 17/P22/Q1/c(ii)

1(c)(ii)	any 2 from: - high melting / boiling / sublimation point - electrical / thermal insulator - hard / rigid - retains strength at high temperature / pressure	2
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58. O/N 16/P22/Q2/b(i)

2(b)(i)	$M_r = \frac{m \times R \times T}{p \times V} \left(= \frac{4.606 \times 8.31 \times 323}{1.68 \times 10^5 \times 1 \times 10^{-3}} \right)$	1
	$= 73.6$	1

59. M/J 16/P21/Q3/b(iii)

(iii)	high melting / boiling point	[1]	[2]
	strong forces (of attraction / between oppositely charged ions) / strong (ionic) bonding	[1]	

60. M/J 15/P21/Q2/a,b

2	(a) (i)	Straight line drawn horizontally from same intercept	[1]	[1]
	(ii)	T_1 because it shows greatest deviation/furthest from ideal	[1]	[1]
	(iii)	reducing T (reduces KE of particles) so intermolecular forces of attraction become more significant	[1]	[1]

(iv)	greatest deviation is at high pressure increasing pressure decreases volume so volume of particles becomes more significant ora	[1] [1]	[2]
(b)	Mass of air = 100×0.00118 = 0.118 g Mass of flask = $47.930 - 0.118$ = 47.812 g Mass of Y = $47.989 - 47.812$ = 0.177 g $pV = nRT = \frac{m}{M_r} RT$ $M_r = \frac{mRT}{pV} = \frac{0.177 \times 8.31 \times 299}{1 \times 10^5 \times 100 \times 10^{-6}}$ = 44.0 (43.979 to 2 or more sf)	[1] [1] [1] [1]	[4]

61. M/J 15/P23/Q1/d

(d) (i)	$pV = \frac{mRT}{M_r}$ $M_r = \frac{mRT}{pV} = \frac{0.275 \times 8.31 \times 298}{100 \times 10^3 \times 200 \times 10^{-6}}$ $M_r = 34.05/34.1$	[1] [1]	[2]
(ii)	(Let % Ne = x so % Ar = 100-x) $\frac{20.2x + 39.9(100 - x)}{100} = 34.05$ % Ne = 29.7	[1]	[1]

62. M/J 15/P23/Q2/c

(c)	(Very) strong electrostatic attraction/ionic bond High charge (density) of cation and anion/ Mg^{2+} and O^{2-}	[1] [1]	[2]
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